SpectraForge

A Physics-Grounded Synthetic Testbed for 4D Fluorescence Hyperspectral Imaging,

and What It Taught Us About Autoencoder Band Selection

Narek Meloyan Narine Sarvazyan
(with an AI research assistant on the training machine)

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Abstract

This is an internal, story-first write-up for supervisors and collaborators. It has two subjects. The first, and the one we care about most, is **SpectraForge**: a physics-grounded generator that turns the equations of fluorescence into synthetic 4D excitation-emission cubes with *perfect ground truth*, built so that the loudest bands are deliberately not the informative ones. We explain what it does, the photophysics and the math behind it, and why it is a tool the 4D fluorescence imaging community can use. The second subject is what we *learned* by running our autoencoder (AE) band-selection idea on it: a debugging story that took a published model from chance to competitive, a significance-tested verdict on where a learned selector genuinely helps (and where classical methods already suffice), and one clean structural result. We keep the account honest, because the value of the benchmark is precisely that it does not let us fool ourselves.

0. The one-page version

- **SpectraForge** renders 4D fluorescence cubes (excitation × emission × height × width) from a parametric fluorophore model obeying Kasha, Vavilov, Franck–Condon and the Stokes shift, with optional inner-filter, reabsorption, FRET, scatter, noise, and instrument clutter. Every cube ships with ground truth, and scenes are constructed so that **band variance is decoupled from class informativeness** (measured correlation ≈ -0.1). That single property is what makes it an *honest* benchmark instead of a trivial one.
- It fills a real gap: labeled real 4D fluorescence data is scarce and has no ground truth, so you cannot tell a selector which bands it *should* have found. SpectraForge can, on demand, at controllable difficulty, with real dye spectra importable from FPbase and an interactive painting GUI.
- Using it, we took a published perturbation-CAE selector **from chance (macro-F1 ≈ 0.33) to competitive**, by fixing four real trainer bugs, a misleading metric, and one architectural flaw (an emission-axis "band collapse").
- The honest, significance-tested verdict: the AE+perturbation selector **matches** the strongest classical blind method (PCA loadings) on realistic data, **significantly beats** it on nonlinear reabsorption (0.449 vs 0.431, 95% CI excludes 0), and **ties** it on FRET. It is a sound, fully-unsupervised selector, not a dominant one.

- The two lessons that generalize: (i) **band selection is not classification** ("informative ⇒ has variance ⇒ PCA finds it"); and (ii) the one place a convolutional model is *structurally necessary* is **spatial-texture** discrimination, where per-pixel selection is provably blind.

1 The problem, and why we built a simulator

A multi-excitation hyperspectral cube illuminates a sample at several excitation wavelengths and, for each, records an emission spectrum across many bands. Every pixel therefore carries an **excitation–emission matrix** (EEM), and a whole scene is a four-dimensional array over (excitation, emission, y, x). This is the modality behind a lot of biomedical and material fluorescence imaging, because different fluorophores, tissue states, or species separate cleanly in the joint excitation–emission domain. The catch is dimensionality: the per-pixel feature vector (excitations $\times$ emission bands) is large and mostly redundant, and only a small subset of bands actually distinguishes the classes. **Band selection** is the task of finding that subset, so a cheaper instrument can acquire only the informative bands, or a classifier can run on a compact feature set.

To develop a band selector you need to know which bands are truly informative, so you can measure whether the selector found them. On real data you never know this. And when people build synthetic data, they almost always make it *too kind*: the discriminative dyes are also the brightest, so "pick the loudest bands" wins and every method looks good. That tells you nothing. What we needed was synthetic data that is (a) physically faithful, (b) equipped with ground truth, and (c) deliberately hard in the way real fluorescence is hard: where the signal you want is *not* the loudest thing in the cube. That is SpectraForge.

2 SpectraForge: turning photophysics into an honest benchmark

2.1 The physics, briefly

A fluorophore absorbs a photon (ground state $S_0 \to S_1, S_2, \ldots$, femtoseconds), tumbles down non-radiatively to the lowest vibrational level of S_1 (picoseconds), and only then emits (nanoseconds). Four textbook consequences shape the EEM, and SpectraForge encodes each:

- **Kasha's rule** — emission leaves from that lowest S_1 level *regardless of how you excited it*. So the emission *shape* does not depend on excitation wavelength, and the EEM factorizes into an excitation profile $\times$ an emission profile $\times$ a concentration. This separability is the basis of PARAFAC, and, as we will see, it *is* Kasha's rule written as algebra.
- **Vavilov's rule** — the quantum yield is, to first order, independent of excitation wavelength. Exciting at 350 or 480 nm changes *how many* molecules emit, not *what colour*: only the amplitude scales.
- **Franck–Condon** — electronic transitions are vertical on the nuclear-coordinate diagram, which sets the broadened, roughly mirror-image band envelopes and an optional vibronic shoulder about $1300\,\mathrm{cm}^{-1}$ to the red (the thing that gives a dye its characteristic shape).
- **Stokes shift** — because emission ends on excited vibrational levels of S_0, the photon is red-shifted from absorption, separating the emission peak from the Rayleigh line.

2.2 The forward model (the actual equations)

Each fluorophore k is a small parametric object: a peak-normalized Gaussian excitation profile $a_k(\lambda_{\text{ex}})$, an area-normalized (optionally red-tailed, vibronic) emission profile $e_k(\lambda_{\text{em}})$, an extinction ε_k, a quantum yield Φ_k (its brightness is the product $\varepsilon_k \Phi_k$), and a per-pixel concentration map $c_k(x, y)$. A single fluorophore's contribution to the cube is

$$S_k(x, y, \lambda_{\text{ex}}, \lambda_{\text{em}}) = \underbrace{c_k(x, y)}_{\text{amount}} \; \underbrace{\varepsilon_k \Phi_k}_{\text{brightness}} \; \underbrace{a_k(\lambda_{\text{ex}})}_{\text{excitation}} \; \underbrace{e_k(\lambda_{\text{em}})}_{\text{emission}}, \tag{1}$$

and the rendered cube is the dilute linear superposition (the linear mixing model, with a global lamp/exposure/power gain g):

$$F(x, y, \lambda_{\text{ex}}, \lambda_{\text{em}}) = g \sum_k S_k. \tag{2}$$

With every optional effect switched off this model is *exactly linear*, and the code checks the invariant $\text{render}(A+B) = \text{render}(A) + \text{render}(B)$. Everything hard about the benchmark is then added back in physically faithful, opt-in increments. Figure 1 shows the dye model: a few dim, overlapping discriminative dyes and a couple of bright, useless nuisances.

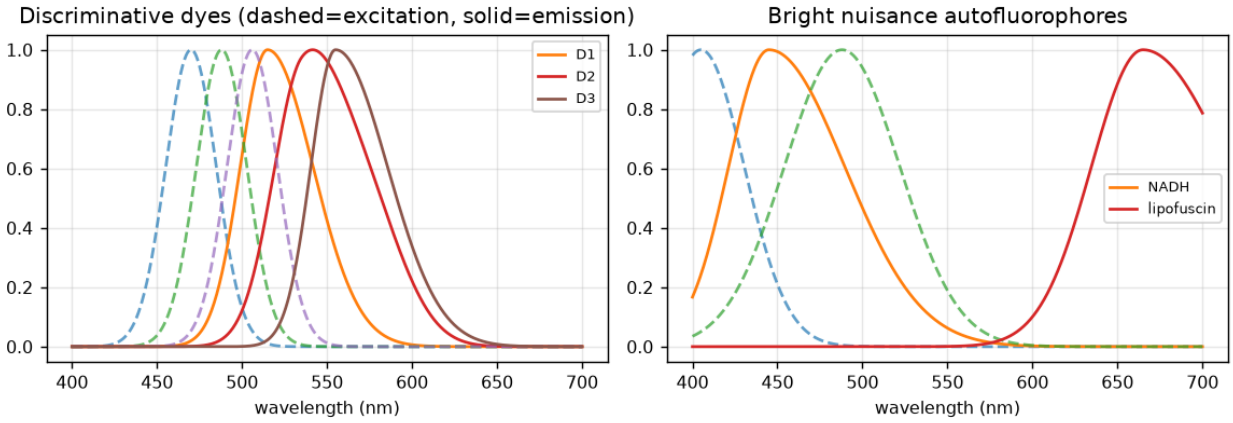


Figure 1: The parametric fluorophore model. Left: three discriminative dyes (dashed excitation, solid emission), modest brightness, overlapping emission, distinct excitation. Right: two bright nuisance autofluorophores (NADH-like, lipofuscin-like) that dominate band variance while carrying no class information.

2.3 The one design decision that matters: variance ≠ informativeness

Here is the heart of it. SpectraForge paints *discriminative* dyes with smooth random concentration fields whose per-pixel argmax *is the class label*; these dyes are deliberately dim and spectrally overlapping, so their discriminative content lives in low-variance band *shape*. Then it paints independent *nuisance* dyes that are bright and spatially varying but class-irrelevant. Real physics decouples variance from class in four independent ways, and the generator reproduces all of them:

1. *high variance, zero information* — Rayleigh and Raman scatter ridges;
2. *high variance, class-irrelevant* — bright autofluorescence nuisances with their own spatial fields;
3. *inflated variance* — shot noise makes bright (possibly redundant) bands look important;

4. *low-variance signal* — the discriminating information sits in dim dyes or in the subtle shape difference between overlapping bands.

The consequence is measurable and is the property that makes the benchmark honest: on a confounded scene the correlation between per-band variance and per-band discriminability is ≈ -0.1, versus strongly positive on naive clean data (Figure 2). A "pick the loudest bands" strategy therefore *fails by construction*, which is exactly what we want, because that is what real fluorescence does.

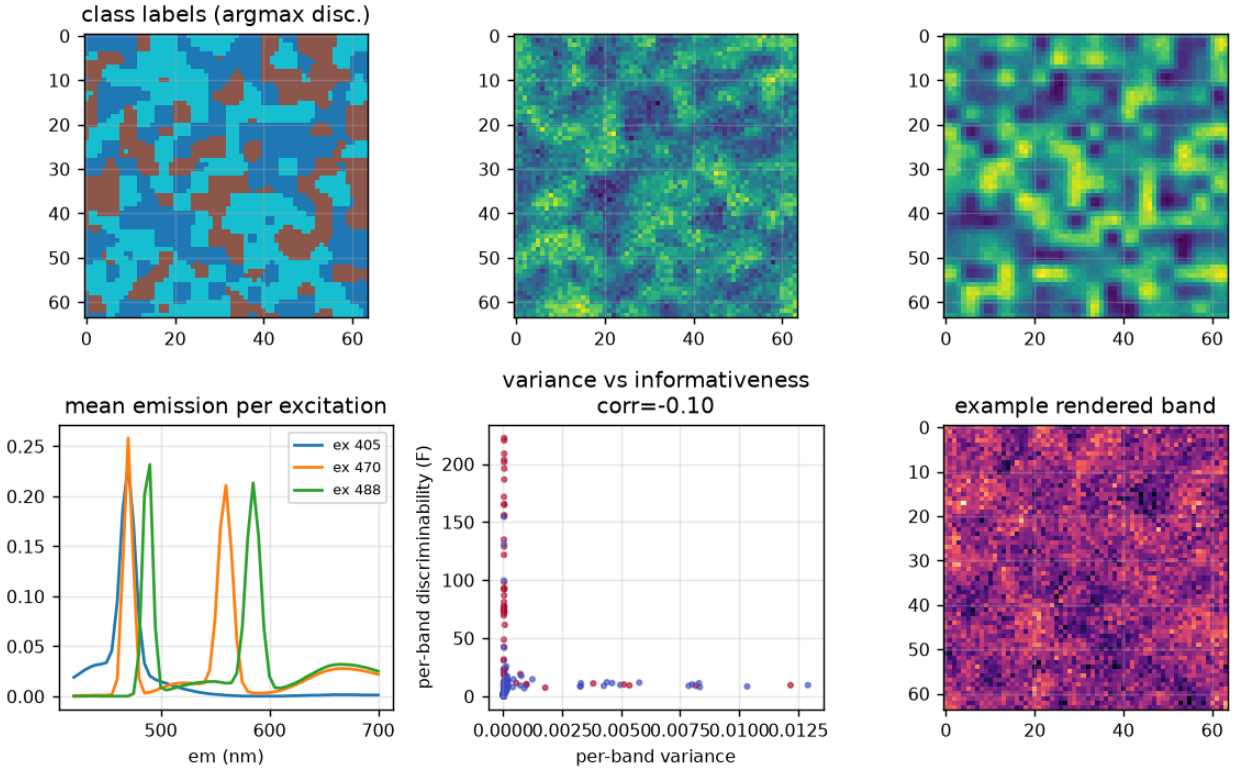


Figure 2: The forward model and the variance-versus-informativeness property. Top: class labels and example rendered bands (a discriminative-dye band, a Rayleigh scatter band, a nuisance band). Bottom: mean emission per excitation; then the scatter of per-band variance against per-band discriminability, whose near-zero (slightly negative) correlation is printed — the loudest bands are not the informative ones; then an example rendered band.

2.4 Making it realistic: scatter, noise, nonlinearity, clutter

On top of the clean cube the generator adds **Rayleigh scatter** (a bright line at $\lambda_{\mathrm{em}} = \lambda_{\mathrm{ex}}$ plus a grating ghost at $2\lambda_{\mathrm{ex}}$) and a **water Raman** line at a fixed wavenumber offset ($\approx 3400\,\mathrm{cm}^{-1}$), each scaled by a per-pixel turbidity field, then **Poisson shot noise** and Gaussian **read noise**. Three optional effects break linearity in concentration, which is where a nonlinear method *could* in principle beat a linear one:

- **Primary inner filter**: local signal attenuated by $\exp(-s\,A_{\mathrm{ex}})$.
- **Reabsorption** (secondary inner filter): emission attenuated by $\exp(-s\,A_{\mathrm{em}}(\lambda))$; because absorption overlaps the blue edge of emission, this *reshapes the band* and apparent-red-shifts the

peak as a function of concentration, relocating discriminative information into band shape (Figure 3).

- **FRET**: for a donor–acceptor pair, a saturating fraction $E = kc_A/(1 + kc_A)$ of donor energy transfers to the acceptor; the donor is quenched and the acceptor sensitized, which encodes dye *co-localization* in a band ratio — a genuinely nonlinear (product/XOR-like) class signal.

Finally, **fixed-pattern clutter** injects many independent class-irrelevant low-rank spatial fields onto random band subsets (the analogue of illumination/detector artifacts). Clutter is high-variance and class-irrelevant, and it turns out to be the dominant real-instrument confound (Figure 4).

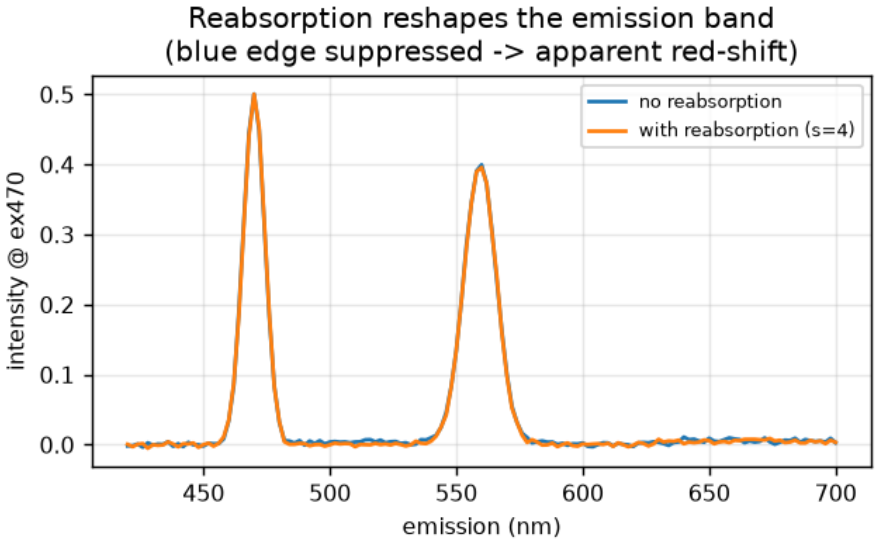


Figure 3: Reabsorption reshapes the band. A bright pixel's emission at ex470 with and without reabsorption (strength 4): the blue edge is suppressed, producing an apparent red-shift — a concentration-dependent, nonlinear reshaping that relocates discriminative information into shape.

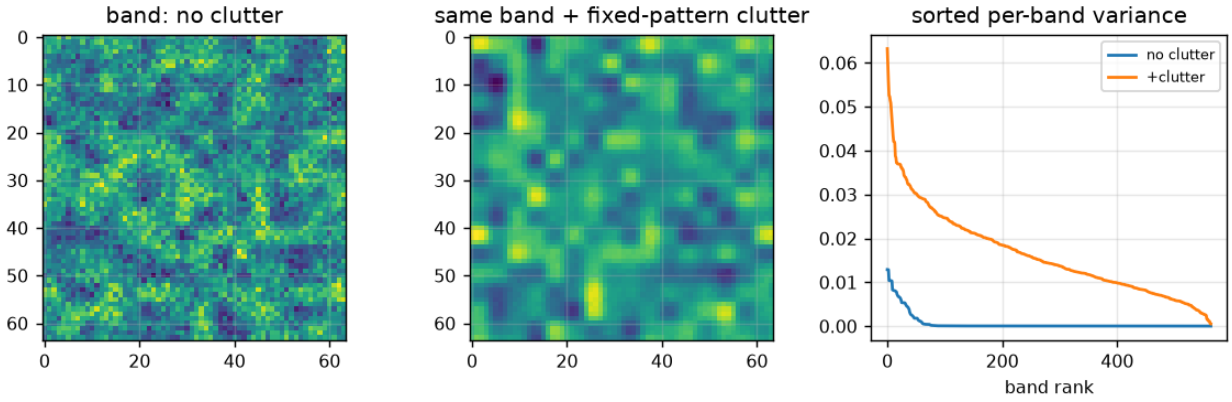


Figure 4: Fixed-pattern clutter. Left/middle: the same band before and after injecting 36 class-irrelevant low-rank spatial clutter modes. Right: sorted per-band variance — clutter raises the variance of many bands without adding any class information, which is exactly why variance- and reconstruction-based selection collapses to chance.

2.5 Regimes, ground truth, and how you actually use it

Dialing these knobs spans a ladder of regimes from one `build_dataset(...)` call: **clean** (variance ≈ informativeness), **realistic** (strong nuisances, scatter, noise), **clutter**, **nonlinear** (reabsorption or FRET/XOR), and **spatial-texture** (class carried by spatial pattern with identical per-pixel spectra). Every render returns a `GroundTruth` sidecar with concentration maps, clean cubes, and each dye's true emission peak, which enables both broad information-retention metrics and a *tight* `peak_recovery` metric (did a selected band land within a tolerance of a true peak? a random selector scores ≈ 0.33). A companion script re-runs every simulation and writes the exact parameters, so figures and numbers are reproducible. The parametric dyes can be swapped for *real measured spectra imported from FPbase*, and there is an interactive PyQt "Forge" workbench where you paint concentration maps, render, and press a button to validate a selection against ground truth.

2.6 Why this is worth having (the point of the tool)

The community lacks an open, physically faithful source of 4D fluorescence cubes with ground truth and controllable difficulty. SpectraForge is that. It lets you (i) benchmark any band-selection, dimensionality-reduction, or classification method against a known answer; (ii) dial difficulty in physically meaningful increments (add scatter, add nuisances, add clutter, turn on nonlinear physics) rather than by fiddling opaque noise levels; and (iii) do all of this while preserving the exact-linear invariant when confounds are off, so you always know what you are testing. And, as the rest of this document shows, it does the thing a good benchmark is supposed to do: it caught a real methodological failure of ours that raw training loss had hidden completely.

3 What we learned running our CAE on it

3.1 The idea, and the crisis

The method we wanted to validate is **autoencoder + latent-perturbation** selection: train an AE to reconstruct the cube; perturb each latent dimension and measure how strongly each input band's reconstruction responds; accumulate those per-band sensitivities into an influence score; keep the top-k (with a diversity constraint). It is appealing because it is fully unsupervised yet, being nonlinear, could in principle catch structure PCA cannot. Run on SpectraForge and scored honestly, the published convolutional version **selected at chance** (macro-F1 ≈ 0.33 on a 3-class scene) and reconstructed almost nothing. Either the pipeline was corrupting pixels, the training was broken, or the architecture could not represent the data. We were warned explicitly that "the loss alone can be deceiving" — and it was.

3.2 Four real bugs

A pipeline-integrity check showed the chunking/batching was lossless to $\sim 10^{-8}$, so the failure was in the model or training. Inspecting the trainer turned up **four genuine bugs**, each independently able to cripple a run:

1. a `ReduceLROnPlateau(verbose=...)` call that *crashes* on modern PyTorch;

2. an `UnboundLocalError` when the loss never improves or is NaN (the best-model path is never assigned, then loaded);
3. a `0.0 * NaN = NaN` sparsity term that poisoned the loss *even when its weight was zero*, so every non-sigmoid-latent run silently trained on NaN and learned nothing;
4. a `scheduler_patience = 5` default that halves the learning rate after five stalled epochs, freezing the model near initialization.

The published default config is byte-identical after the fixes; we only removed the failure modes.

3.3 The metric was lying too

Our own first verdict — "$R^2 < 0$, reconstructs nothing" — used per-band-averaged R^2, which is dominated by the ~ 40 of 57 off-peak noise bands that *no* model can reconstruct. Switching to signal-band correlation, and measuring the achievable ceiling with a per-pixel MLP control (≈ 0.49, not 1.0), reframed every later comparison. Lesson: on sparse spectra, average reconstruction error is not a proxy for whether the model captured the signal.

3.4 The architectural flaw: the band collapse

With the bugs fixed and the honest metric in hand, the published CAE *still* reconstructed nothing on multi-fluorophore data (signal correlation ≈ 0.001) yet worked on a single-dye scene (≈ 0.34). That dissociation localized the fault exactly: the emission-axis `adaptive_avg_pool3d` **averages over emission**, so it cannot represent different spatial patterns at different emission bands — which is precisely what distinguishes multiple fluorophores. Removing the collapse, and nothing else about the idea, raised reconstruction about $100\times$ ($0.001 \rightarrow 0.104$) and took selection **from chance** 0.331 **to** 0.414**, about** 85% **of the labels-using oracle (**0.486**)**. So the answer to "is it a bug or the architecture?" was *both*.

3.5 Recovery, and a swarm

A repeatedly-confirmed finding shaped the search: **reconstruction fidelity is anti-correlated with selection quality** — the selection knobs matter more than the last bit of reconstruction. Tuning the perturbation (PCA-ranked latent dims, standard-deviation perturbation, ~ 80 accumulated dims) lifted selection to 0.438 (90% of oracle); a per-pixel collapse-*bottleneck* conv-AE reached $\mathbf{0.473 \pm 0.018}$ on clean, matching PCA loadings (≈ 0.48, 97% of oracle). A flexible-architecture swarm over depth, width, latent size, activation, masking/denoising, and MLP-vs-1D-conv backbones found that masking/denoising is the dominant lever on clean data (0.50–0.51, exceeding PCA), and a conv-over-bands backbone at modest depth is best on realistic data (0.464); depth and width beyond that gave nothing, confirming a ceiling rather than a tuning shortfall.

3.6 The honest, significance-tested verdict

To keep ourselves honest we ran a full battery: five metric families over eight selectors, several regimes, six seeds, with bootstrap confidence intervals and, crucially, a **random** baseline and the

full spectrum as controls. The headline comparison of the best AE against the strongest classical blind selector (PCA loadings) is Table 1.

Table 1: Significance battery: AE (best of family) vs PCA loadings, best-of-panel macro-F1, six seeds. Positive margin favours the AE.

regime	AE	PCA-load	margin	95% CI
clean	0.590	**0.635**	-0.046	$[-0.079, -0.019]$
realistic	0.527	0.531	-0.004	tie
reabsorption	**0.449**	0.431	$+0.018$	$[+0.001, +0.029]$
FRET	0.635	0.642	-0.007	tie

Read plainly: PCA loadings is the strongest blind selector overall; the AE is **competitive, not dominant**. It ties on realistic and FRET, it *significantly* beats PCA on nonlinear reabsorption (the confidence interval excludes zero, Cohen $d \approx 0.94$), and PCA wins on clean, low-noise data where the informative bands are also the loudest. The AE's clean advantage seen in the swarm is real but **noise-dependent** — its denoising only helps when there is noise to remove.

3.7 Trying to beat PCA "by far": the nonlinear tests

An AE *should* win where the structure is nonlinear. We tested this directly. Under physically-correct reabsorption, with an evaluation that credits nonlinear shape (a classifier panel and a nonlinear-only "gap" metric), the AE's subset carries nonlinear-only information ($+0.030$) that PCA's does not (-0.021), and the margin crosses from negative to positive as the regime becomes more nonlinear — the direction of the hypothesis is confirmed. But the margin is small ($\lesssim 0.03$) and did not reproduce at three independent seeds, so "by far more" is not supported on this data class. The strongest honest nonlinear regime is FRET on an XOR co-localization scene, where even a marginal mutual-information oracle fails (XOR is invisible per band). There the AE and PCA are **tied** (0.633 vs 0.641, within noise), and, tellingly, the AE's nonlinear-only gap ($+0.092$) *equals* PCA's ($+0.091$). It *captures* the nonlinear structure; it just does not *select* better. The reason is the single most useful thing we learned:

> **Band selection is not classification.** Informative $\Rightarrow$ has variance $\Rightarrow$ PCA selects it. The discriminative bands carry concentration-driven variance, so a linear selector finds them; the nonlinearity (XOR, a FRET ratio) is then resolved by the *downstream* classifier, not by the selector. A nonlinear classification task does not imply a nonlinear selection task, so the AE's manifold advantage buys nothing extra at the selection step.

3.8 Where the benchmark says to stop

Because we always ran the two controls, SpectraForge also told us where unsupervised selection stops helping (Figure 5). Under realistic fixed-pattern clutter, **blind selection (AE and PCA alike) does not beat a random subset** — only label-aware selection does — because clutter is high-variance but class-irrelevant, so every variance-driven criterion ranks clutter bands about randomly with respect to class. Across label budgets, **selection never beats the full spectrum**: its value is compression, not accuracy (both blind selectors reach $\sim 95\%$ of full at a large budget,

while labels compress about 2.7× better). And both blind selectors are **unstable** band by band (Jaccard ≈ 0.1–0.2), which argues for selecting spectral regions or consensus sets rather than exact bands. These are findings of the benchmark, and they are useful: they tell a practitioner when to bother selecting bands at all.

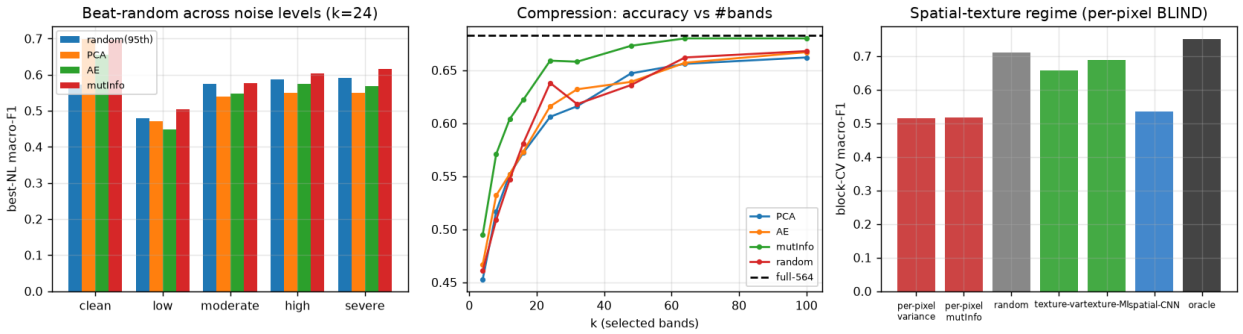


Figure 5: What the controls reveal. Left: beat-random across five noise levels — blind PCA/AE beat the random 95th percentile only on clean data; under any clutter only supervised mutual information does. Middle: the compression curve — AE tracks PCA at every budget, both reaching ∼ 95% of full, while labels compress ∼ 2.7× better. Right: the spatial-texture regime under honest block cross-validation — per-pixel selectors are the *worst* (below random), while a spatial model and the oracle succeed.

3.9 The one clean structural win

There is exactly one regime where the convolutional idea is *structurally necessary*. When the class is carried by **spatial texture**, so every class has the same per-pixel spectrum, per-pixel band selection is **provably blind**: under honest spatial-block cross-validation the per-pixel selectors are the worst (0.52–0.55, below the random 0.674), while a spatial model with the discriminative bands reaches the oracle (0.78). This vindicates the original spatial-CAE intuition — but for *spatial classification*, not for per-pixel spectral band selection.

4 What it all means

Three mechanisms explain every result above.

1. **Selection ≠ classification.** The AE's nonlinearity helps a classifier, not the choice of bands.
2. **Unsupervised ≡ variance.** Reconstruction, variance, and PCA loadings all rank by variance, which under clutter or nuisances is decoupled from class — so they behave like random.
3. **Extractable ⇔ variance-prominent.** A band strong enough to be useful is high-variance (simple methods find it); a band that is not variance-prominent is too weak to extract. There is no gap for a learned *selector* to fill — except genuinely spatial information, which is invisible to per-pixel methods.

5 Recommendations

- For per-pixel band selection on realistic data, use **supervised** selection (marginal mutual information for clutter/linear structure, random-forest importance for nonlinear), and **always** benchmark against random and full-data. If a method cannot beat a random subset, it adds nothing.
- Treat band selection as a **compression** tool. If you have the labels and the budget, keep the full spectrum.
- Select **spectral regions or consensus sets**, not exact bands — no selector is stable under clutter.
- Use a **convolutional/spatial** model where discrimination has spatial structure; that is its real, provable value.
- Validate on **real data** with the same controls. This is the open gate.

6 Open gate: real data

The decisive next step is to run the winning configuration on the actual Lichens and Collagen cubes, scored with the same panel and the same random/full controls. Real photophysics may place informative structure where it is *not* variance-prominent — the one setting in which the synthetic conclusion (AE $\approx$ PCA for selection) could genuinely change. SpectraForge exists precisely so that, whatever real data says, we will be able to tell whether it is a real effect or another way of fooling ourselves.

References

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Appendix: exact configurations

Emission 420–700 nm (2 or 5 nm step $\rightarrow$ 141 or 57 bands), excitations $\{405, 470, 488, 506\}$ nm, scene 64×64. Realistic regime: `nuisance_amp=1.2, photon_scale=2000, read_sigma=0.005`.

Clutter: `modes=36, amp=3.3`. Reabsorption: `strength=4`. Every figure is produced by re-running the simulation via `reports/paper_figures.py`, which writes the exact parameters to `paper/figures/params.json`; the full campaign log is `docs/spectraforge/OVERNIGHT-LOG.md` and the significance battery is `reports/exp_records/metric_suite.csv`.